

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA0307	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
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Section / Batch:	B.E-CSE-AI	Date:	

Code of Conduct

I Shaik Nuha(192472242) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sk.Nuha

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus:

"Data science is powerful, data science drives innovation, data science is evolving."

The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences.

The following statistics are obtained from the corpus:

- $C(\text{data}) = 3$
- $C(\text{data science}) = 3$
- $C(\text{science is}) = 2$
- $C(\text{science drives}) = 1$
- $C(\text{science drives innovation}) = 1$

The interpolation weights are:

- $\lambda_1 = 0.5$ (Trigram)
- $\lambda_2 = 0.3$ (Bigram)
- $\lambda_3 = 0.2$ (Unigram)

The prediction probabilities after the word "science" are:

- $P(\text{is}) = 0.66$
- $P(\text{drives}) = 0.33$

Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability $P(\text{science} | \text{data})$. Interpret how this probability influences next-word prediction in a mobile keyboard application.
 2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.
 3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
 4. Calculate the Entropy of the distribution $P(\text{is})=0.66$ and $P(\text{drives})=0.33$. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.
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CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.

Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.

Given:

- $P(\text{book} \mid \text{VB}) = 0.6$
- $P(\text{book} \mid \text{NN}) = 0.4$
- $P(\text{Start} \rightarrow \text{VB}) = 0.5$
- $P(\text{Start} \rightarrow \text{NN}) = 0.5$

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
 2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
 3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
 4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.
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CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
 - economic (120 occurrences)
 - growth (450 occurrences)
 - increases (210 occurrences)
 - employment (380 occurrences)

Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.
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Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1.Smart Mobile Keyboard Prediction System

Question 1: Compute the MLE of the Bigram Probability

Formula

$$P(\text{Science}|\text{data}) = C(\text{data}) / C(\text{data science})$$

Given

- $C(\text{data}) = 3$
- $C(\text{data science}) = 3$

Calculation

$$P(\text{science}|\text{data}) = 3/3 = 1.0$$

Answer

$P(\text{Science} \text{data}) = 1.0$

Interpretation

- The word science always follows data in the training corpus.
- Therefore, the keyboard predicts science with 100% probability after the user types data.
- This improves typing speed and prediction accuracy.

Question 2: Backoff Model

Input Sequence: data science improves

The trigram "data science improves" does not occur in the corpus.

Step 1: Trigram

$$P(\text{improves} | \text{data science}) = 0$$

Since the trigram is unseen, back off to the bigram.

Step 2: Bigram

$$P(\text{improves} | \text{science}) = 0$$

This bigram is also unseen.

Step 3: Unigram

Since improves is unseen in higher-order models, use its unigram probability.

$$P(\text{improves}) = N/C(\text{improves})$$

Because improves is not present in the corpus,

$P(\text{improves}) \approx 0$

(or a very small smoothing probability in practice).

Answer

The Backoff Model estimates the probability using the lower-order unigram when trigram and bigram probabilities are unavailable.

Why Backoff is Important

- Handles unseen word sequences.
- Prevents zero probabilities.
- Improves prediction coverage.
- Enables robust real-time text prediction.
- Enhances user experience.

Question 3: Deleted Interpolation

Formula

$$P = \lambda_1 P_{\text{tri}} + \lambda_2 P_{\text{bi}} + \lambda_3 P_{\text{uni}}$$

Given

- $\lambda_1 = 0.5$
- $\lambda_2 = 0.3$
- $\lambda_3 = 0.2$

For the sequence data science is

Trigram probability

$$P_{\text{tri}} = 2/3 = 0.667$$

Bigram probability

$$P_{\text{bi}} = 2/3 = 0.667$$

Unigram probability

$$P_{\text{uni}} = 2/11 = 0.182$$

Calculation

$$\begin{aligned} P &= (0.5)(0.667) + (0.3)(0.667) + (0.2)(0.182) \\ &= 0.3335 + 0.2001 + 0.0364 \end{aligned}$$

$$\boxed{P = 0.57}$$

Interpretation

Deleted Interpolation combines trigram, bigram, and unigram probabilities, ensuring reliable predictions while reducing data sparsity problems.

Question 4: Entropy Calculation

Formula

$$H = -\sum P(x) \log_2 P(x)$$

Given

- $P(\text{is}) = 0.66$
- $P(\text{drives}) = 0.33$

Calculation

$$H = -(0.66 \log_2 0.66 + 0.33 \log_2 0.33)$$

$$H = -(0.66 \times -0.599 + 0.33 \times -1.599)$$

$$= 0.395 + 0.528$$

$$H \approx 0.92 \text{ bits}$$

Interpretation

- Low entropy indicates higher confidence in predictions.
- Since is has a much higher probability than drives, the keyboard is confident about suggesting is.
- Entropy helps measure uncertainty and prediction confidence.

CASE STUDY 2: AI-Powered Customer Support Chatbot

Question 1: Penn Treebank POS Tagging

Sentence 1: Book a flight ticket now.

Word	POS Tag	Meaning
Book	VB	Verb
a	DT	Determiner
flight	NN	Noun
ticket	NN	Noun
now	RB	Adverb

Sentence 2: This book is interesting.

Word	POS Tag	Meaning
This	DT	Determiner
book	NN	Noun
is	VBZ	Verb
interesting	JJ	Adjective

Justification

- In "Book a flight ticket now", Book is an action, so it is tagged as VB.
- In "This book is interesting", book refers to an object, so it is tagged as NN.

Question 2: HMM Probability

Formula

$$P(VB)=P(\text{Start} \rightarrow VB) \times P(\text{book}|VB)$$

Given

$$P(\text{Start} \rightarrow VB)=0.5$$

$$P(\text{book}|VB)=0.6$$

Calculation

$$P(VB)=0.5 \times 0.6$$

$$P(VB)=0.30$$

Similarly,

$$P(NN)=0.5 \times 0.4=0.20$$

Since

$$0.30 > 0.20$$

the HMM chooses VB.

Answer

The probabilistic model favors Verb (VB) because its probability (0.30) is higher than the noun probability (0.20).

Question 3: Rule-Based Tagging vs HMM Tagging

Rule-Based Tagging	HMM Tagging
Uses predefined grammar rules	Uses statistical probabilities
Simple to implement	Learns from training data
Poor handling of ambiguity	Resolves ambiguity effectively
Less scalable	Highly scalable
Lower accuracy	Higher accuracy

Recommendation

HMM Tagging is recommended because it:

- Handles ambiguous words accurately.
- Adapts to different contexts.
- Works well with large datasets.
- Produces better chatbot responses.
- Improves overall tagging accuracy.

Question 4: Role of Standardized POS Tagsets

Standardized POS tagsets such as the Penn Treebank Tagset improve chatbot performance by:

- Providing consistent grammatical annotation.
- Improving intent detection.
- Enhancing response generation.
- Resolving lexical ambiguity.
- Supporting accurate syntactic parsing.

- Increasing NLP model accuracy.
- Improving overall user interaction.

Conclusion

Using Penn Treebank POS Tagsets together with Hidden Markov Models (HMMs) enables chatbots to identify the correct grammatical role of words, resolve ambiguity, and generate more accurate and context-aware responses.